

# ZARI.ai - Master Thesis Defense Preparation Report

3-Crop Plant Disease Diagnostics, EDL Uncertainty, Multilingual RAG & IPM Advisory Engine

## 1. Executive Defense Summary & Core Metrics

ZARI.ai is an end-to-end plant disease diagnostic and context-aware recommendation platform for Tomato, Potato, and Bell Pepper crops in Pakistan.

\* Master Dataset: 49,805 images (39,834 Train, 4,978 Val, 4,993 Test) across 26 canonical classes.

\* Final System

## 2. Technology Stack & Framework Components

\* Core Deep Learning: PyTorch 2.x, timm 1.0.28, torchvision, OpenCV 4.x, PIL, NumPy, Pandas.

\* Model Backb

## 3. Models Trained & Comprehensive Evaluation Matrix

### A. Model A (Crop Router Classifier)

\* Backbone: E

### B. Model B (Crop-Specific EDL Disease Classifiers)

\* Tomato Cl  
best\_model\_b

### C. Swin-Tiny Vision Transformer (ViT) Comparative Evaluation

\* Swin-Tiny Te

### D. Knowledge Distillation Experiment (Swin-Tiny Teacher -> EfficientNet Student)

\* Hyperparam

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4. ChromaDB Vector Store & Embedding Engine

\* Collection Name: zari\_3crop\_treatment\_kb

\* Storage Engine: DuckDB

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5. Complete Model Parameter Breakdown

\* Model A Crop Router (EfficientNetV2-B2): 7,705,221 parameters (7.71 M)

\* Model B Tomato Classifier (ViT-B/16): 86M parameters

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6. Real CUDA System Latency Breakdown

| Pipeline Stage | Mean Latency (ms) | Median Latency (ms) | P90 Latency (ms) | Percentage of System Time |

| :---- | :---- |

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7. Defense Panel Expected Q&A Preparation

Q1: Why did you use Evidential Deep Learning (EDL) instead of standard Softmax?

A: Softmax for classification tasks parameterizes an explicit evidence space.

Q2: Why select paraphrase-multilingual-MiniLM-L12-v2 over larger LLM embeddings?

A: MiniLM-L12-v2 provides high-quality embeddings at a much smaller size, reducing inference time and memory footprint.

Q3: Why was Swin-Tiny Vision Transformer rejected for production?

A: Swin-Tiny achieved high accuracy but required a large number of parameters and complex hierarchical structure, making it difficult to deploy in resource-constrained environments. EfficientNetV2 was chosen for its better trade-off between accuracy and efficiency.

Q4: How does SAM2 Leaf Masking compute disease severity?

A: Meta SAM2 uses a promptable segmentation model to identify and mask diseased leaf regions from binary segmentation maps. The area of the masked regions is then used to measure exact disease severity.

Q5: How is IPM compliance enforced in chemical recommendations?

A: The RAG system cross-references chemical recommendations with a database of IPM regulations and restrictions to ensure compliance.

strictly restricted  
days.

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## 8. Final Thesis Summary & Deliverables

\* PDF Defense Report: ml\_pipeline/reports/ZARI\_THESIS\_DEFENSE\_MASTER\_REPORT.pdf

\* Markdown D